PROJECT PRESENTATION

Lab Equipment Management System

LEMS

A web-based platform for equipment booking, maintenance reporting and technical support.

Dmytro Varich · Daniil Sputai · Artem Polianetskyi · Myroslav Yatsenko

TUKE / IAI



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One system. Two user roles.

LEMS centralizes equipment access, booking requests and malfunction reporting.

WHAT THE PLATFORM DOES

Manage laboratory equipment in one place.

Students can browse equipment, request bookings, track their requests and report problems. Administrators manage equipment, review bookings and handle reported malfunctions.

01

Student

Browse equipment
Submit booking requests
Report issues

02

Administrator

Manage equipment
Approve or reject bookings
Handle reported issues

A

BOOK

Equipment reservations

B

TRACK

Booking status

C

REPORT

Malfunction reports

The process model behind LEMS

BPMN describes the collaboration between equipment management and technical support.

60+

BPMN activities

Across multiple
business processes

01 **Equipment Management System**

Students · Admin · System

02 **Technical Support Center**

Issue handling and support

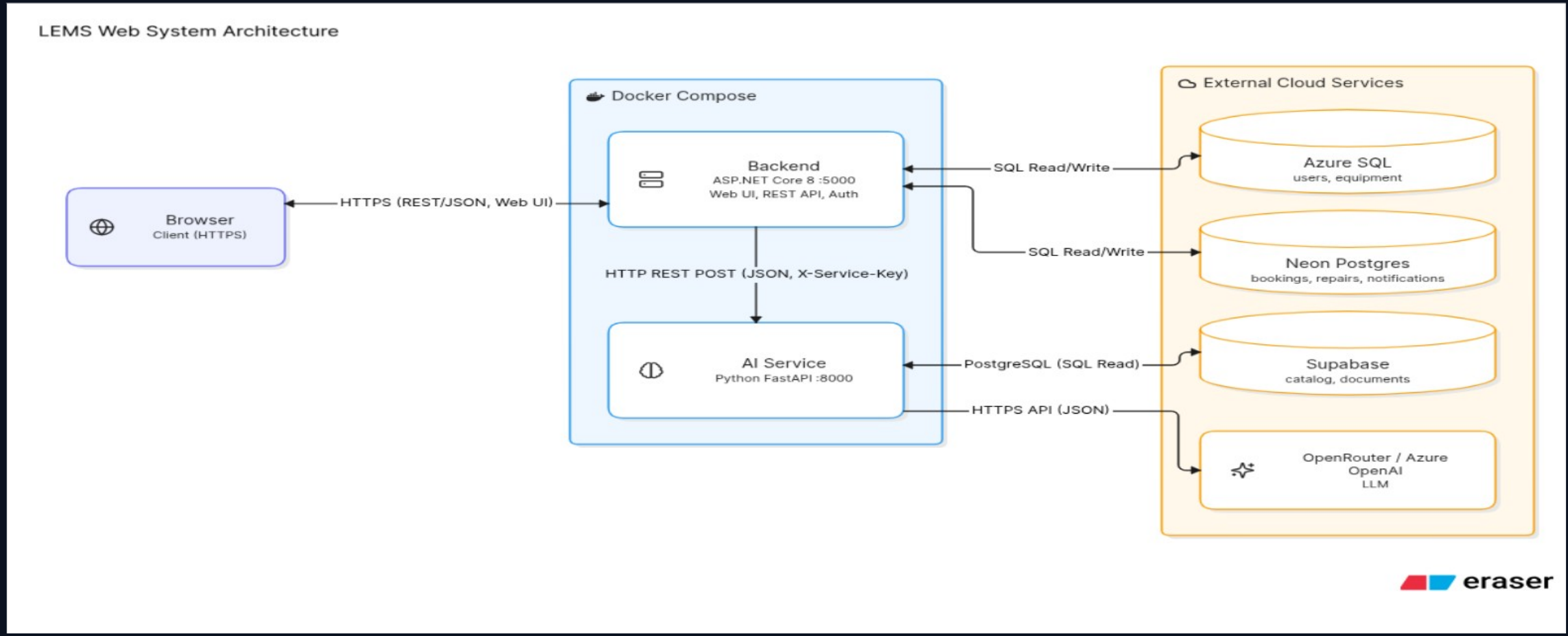
03 **Equipment Maintenance**

Maintenance subprocess

FULL BPMN DIAGRAM → github.com/Archi816/AIS_PU-project

A modular web-system architecture

The backend coordinates the browser, AI service and external data / cloud services.



From access to resolution

A typical student journey through the main LEMS capabilities.

01

Sign in

Users log in as a student or administrator. The role determines which features are available.

02

Book equipment

Students browse available equipment, choose a date and time slot, and submit a booking request. Admins approve or reject it.

03

Report an issue

Students submit a malfunction report. Administrators review the problem and coordinate the next steps.

04

AI assistance

The AI service analyses the issue using the description and available knowledge-base documentation, then suggests troubleshooting steps.

Thank you.

Questions & discussion

Project repository

github.com/Archi816/AIS_PU-project

